Lecture 1:
Introduction to AI and Agents

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Agenda

- ➢Introduction and overview of the module
- ➢Introduction to Artificial Intelligence
- ➢Applications of Artificial Intelligence
- ➢Agents and environment
- ➢Reflex, goal-based agents
- ➢Agentic AI

Module Overview

- Semester Long Module
- Fully assessed by coursework
- 2 courseworks
  - CW1 (25%) – Research work on a chosen AI topic
  - CW2 (75%) – Development of an AI system based on topic chosen in CW1
- Students will be required to present their work in form of a VIVA and Presentation

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Learning Outcomes

- LO1- Gain knowledge and understanding of artificial intelligence
- LO2 - Be able to work effectively in teams to design an intelligent system
- LO3 - Demonstrate ability to communicate effectively on the topic of AI
- LO4 – Be able to develop and evaluate a simple AI prototype from a brief

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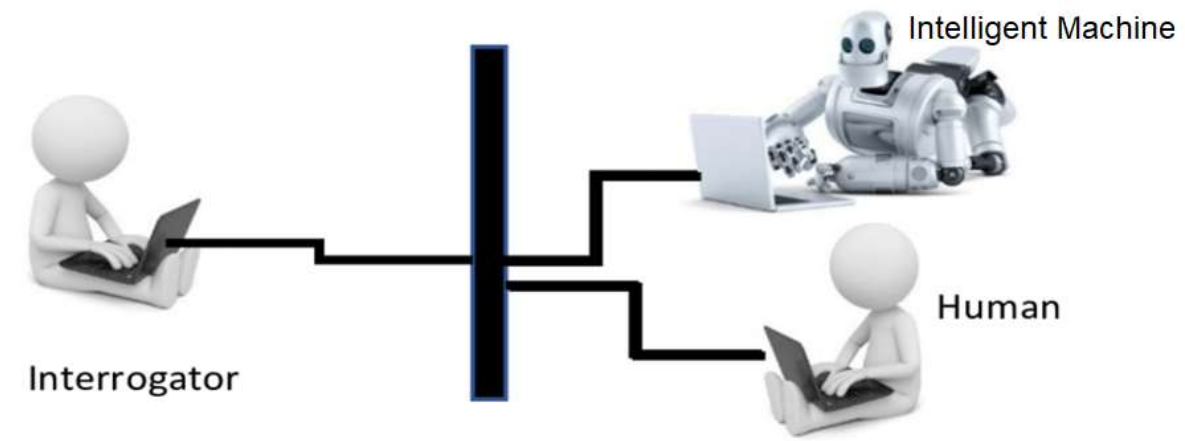
AI: Early phase

John McCarthy (1927–2011), an American computer scientist and cognitive scientist, often hailed as the "**father of artificial intelligence**" (AI)

Turing Test



Chess (1997): Kasparov vs IBM Deep Blue
Jeopardy! (2011): Humans vs. IBM Watson

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Artificial Intelligence?

"**Intelligence**: The ability to learn and solve problems" - *Webster's Dictionary.*

"**Artificial intelligence** (AI) is the intelligence exhibited by machines or software" - *Wikipedia.*

"The science and engineering of making intelligent machines" – ***John McCarthy***.

"The study and design of *intelligent agents*, where an intelligent agent is a system that *perceives its environment* and takes actions that *maximize its chances of success*." - *Russel and Norvig AI book.*

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Types of AI

- **Weak AI (Narrow AI)**: Performs a specific task (e.g. chatGPT, Siri ,Google Assistant).
- **Strong AI (General AI)**: Machines with the ability to perform any intellectual task that a human can do (future concept).
- **Super AI**: A level where AI surpasses human intelligence.

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Current scenario of AI

- From Narrow AI to Generalization
- AI as a Collaborator, not just a Tool
- Democratization of AI
- Ethical & Responsible AI
- AI & Human Augmentation

Recent Outlook of AI

- Generative AI Dominance
- Rise of Multimodality
- AI Agents & Autonomy
- AI in Scientific Discovery
- Regulation & Governance
- Sustainability & Green AI

Future Trends

- Personalized AI
- Hybrid Intelligence

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What is AI and What is not AI?

AI	Not AI
Customer support chatbot user asks something new, the AI can attempt to answer based on its training and learning from past conversations	**Simple rule-based chatbot** Answers **predefined questions**: "What is the price of Coffee?“ ➡ "The price is 120.“ "How much does Coffee cost?" ➡ the bot may not recognize it
AI-based spam filter (e.g., Gmail’s spam detection) Learns from large datasets of spam and non-spam emails. improve by analyzing new emails and recognizing patterns **Adaptability:** It gets better at identifying spam as more emails are processed	**A rule-based spam filter.** The system uses set of predefined rules, conditions "If an email contains the word ‘lottery’” ➡ “mark it as spam.“ **Limitation**: If spammers start using a different word or tactic, the filter won’t catch the spam

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State-of-the-art Applications

- Speech recognition
- Autonomous planning and
- Scheduling
- Financial forecasting
- Game playing, video games
- Spam fighting
- Logistics planning
- Robotics (household, surgery, navigation)
- Machine translation
- Information extraction
- Automatic assembly
- Sentiment analysis
- Fraud detection
- Recommendation systems
- Web search engines
- Autonomous cars
- Energy optimization
- Question answering systems
- Social network analysis
- Medical diagnosis, imaging
- Route finding
- Traveling salesperson
- Document summarization
- Transportation/scheduling
- Computer animation
- *Many more !!*

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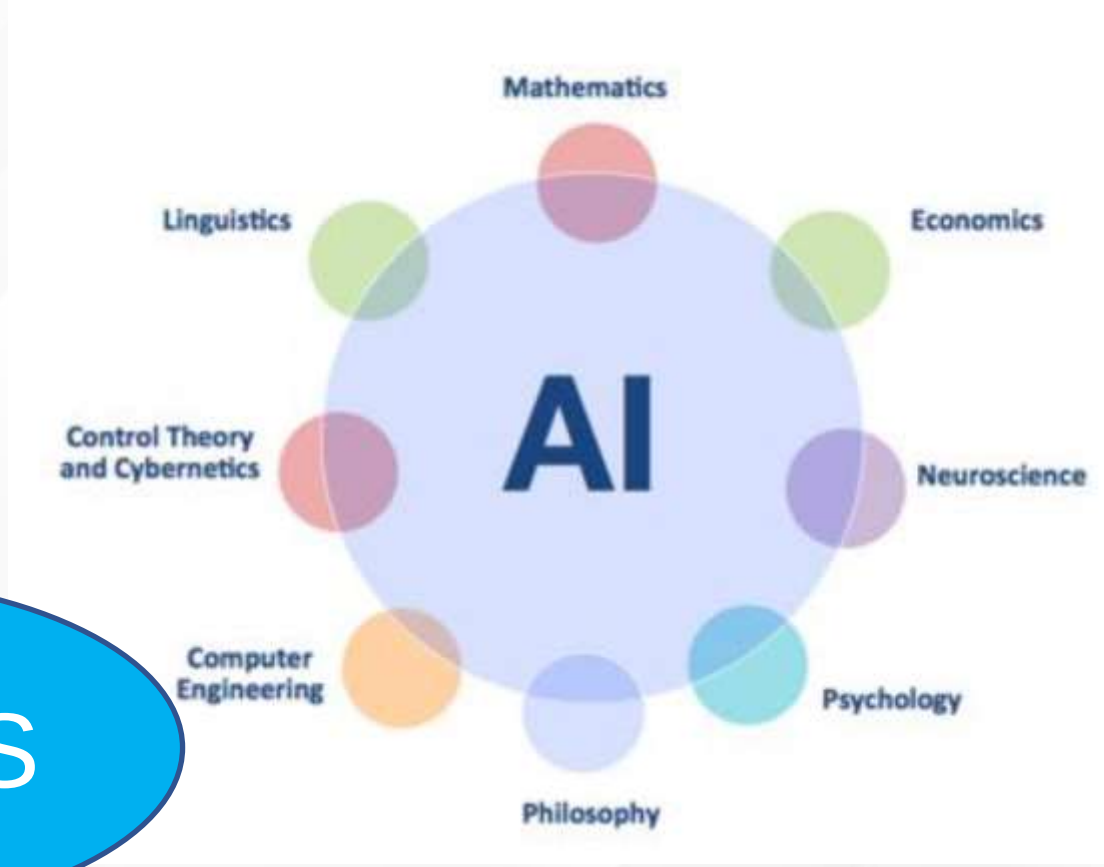
Foundation of AI



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History of AI

- 1940-1950: Gestation of AI
  - Turing's Computing Machinery and Intelligence
- 1950-1970: Early enthusiasm, great expectations
  - Birth of AI @ Dartmouth meeting 1956.
- 1970-1990: Knowledge-based AI
  - Expert systems, AI becomes an industry
  - AI winter
- 1990-present: Scientific approaches
  - Neural Networks: le retour (the return)
  - The emergence of intelligent agents
  - AI becomes "scientific", use of probability to model uncertainty
  - AI Spring!

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Agent or not?

Agents vs plants and rocks "It *reacts* to sunlight, so is it an agent?"

Not an agent. It doesn't make *decisions*.

Mom shouting, "Study well!"

"She perceives your marks and future and acts accordingly!"

In a nutshell, a human agent with goals and feedback

"Ball goes zoom → dog goes zoomier"

Senses (eyes), acts (runs), has a goal (fetch)

A pizza ordering chatbot

"You type 'extra cheese', it says 'Sure thing!'

Software agent – senses input, performs action

Delivery drones "Sees a cat blocking your door and still delivers your order!"

Robotic agent with navigation and obstacle avoidance

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Agents

- *What is the agent doing? What does it know? What does it want to achieve?*
- "Understanding agents and environments is like learning the alphabet before writing a story—it's where all intelligent action begins."
- An **agent** is anything that **perceives** its environment through sensors and **acts** upon that environment through actuators.
- An agent is an entity that perceives its environment and takes actions to achieve goals.

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Types of Environments

Environment	Detail	Example
Fully / Partially Observable	All relevant info is available via sensors	Chess (Fully), Driving (Partially)
Deterministic / Stochastic	Future states are predictable vs random	Calculator (Deterministic), Stock market (Stochastic)
Episodic / Sequential	Decisions are independent or linked over time	Image classification (Episodic), Robot navigation (Sequent
Static / Dynamic	Whether the environment changes while the agent is deliberating	Crossword puzzle (Static) Real-world driving (Dynamic)
Discrete / Continuous	Limited number of percepts and actions or not	Board games (Discrete) Real-time simulations (Continuous)
Single-agent / Multi-agent	One or more decision-makers	Solitaire (Single), Soccer (Multi-agent)

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A Broken Reflex Agent

- A reflex-based vacuum cleaner that always turns left.
- Walk like a vacuum that always turns left and crashes into the same chair.
  - ***"Is this agent rational?"***
  - ***"Is it acting autonomously?"***
  - ***"How could we improve it?“***
- Takeaway: Reflex agents need good condition-action rules, and ideally some model of the world.

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Agent Explore

Agent Idea	Element	Angle
A dating app matchmaker AI	“Only matches based on favorite pizza toppings”	Bad performance measure
A smart fridge that locks at midnight	“Because the agent thinks you're on a diet”	Sensors (time), Actuators (lock), Wrong goals
A “smart” umbrella	“Opens whenever anyone near it sneezes”	Bad environment perception logic

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PEAS Framework

Used to define an agent **formally** in terms of its task environment

Performance: What defines success for the agent

Environment: The context or world the agent operates in

Actuators: What the agent uses to act on the environment

Sensors: What the agent uses to perceive the environment

Example: Self-Driving Car

PEAS Component Example

Performance Measure:
Safety, speed, fuel efficiency, and passenger comfort

Environment:
Roads, traffic, pedestrians, weather

Actuators:
Steering, brakes, throttle, indicators

Sensors:
Cameras, radar, GPS, LIDAR

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Rationality and Autonomy

Rational Agent

- A **rational agent** is one that **does the right thing**, based on:
- The **performance measure**
- Its **percept history**
- Its **knowledge** about the environment
- **Rational ≠ Omniscient**
  Rational agents aim to make the **best possible decision** with the information available.

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Rational Intelligent Agents

- An agent *perceives the environment* and *acts upon* that environment to *achieve some task*.
- We care specifically about *rational agents*.
- Rational agents act based on or in *accordance with reason or logic*.
- Rationality is relative to how to act to maximize a *performance measure*.
- AI aims to design the best agents (programs) that achieve the best performance given the computational limitations.

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Autonomy

- Degree to which an agent relies on its own experience rather than built-in knowledge.
- Highly autonomous agents learn and adapt over time.

Example:

- A GPS app that always adapts to new traffic patterns is more autonomous than one that uses fixed routes.

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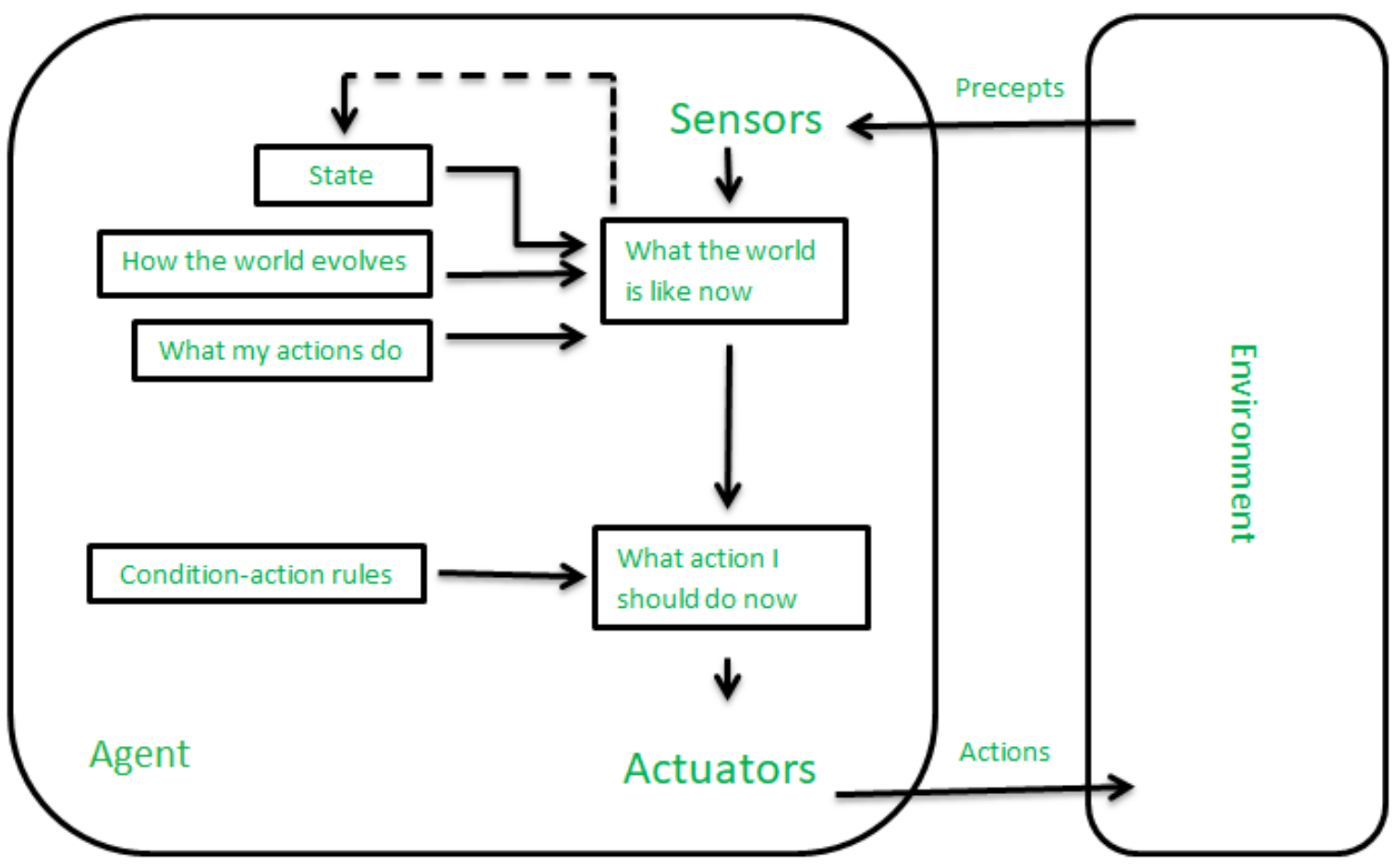
Agent Programs



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Simple Reflex Agents

Act only on the **current percept**, not based on history.

Structure:

- Percept → Condition-action rule → Action
- **No memory, no understanding of world state**

Eg: If there's dirt, then suck it.

Fails in partially observable environments

Cannot handle unseen situations or change strategy

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Model-Based Reflex Agents

- Use an **internal state** (model of the world) to keep track of unobserved parts.
- Percept + internal state → Update state → Apply rule → Action
- Keep track of which rooms were already cleaned in case you can't see them.
- **Improvement**: Works in partially observable environments.

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Goal-Based Agents

Decide actions based on **goals** rather than fixed rules.

Behavior:

•Evaluate different actions to choose one that moves it closer to a goal.

•Needs **search** and sometimes **planning**.

Example:

Self-driving car: “I want to reach my destination safely and quickly.”

Key Benefit: More flexible, adaptive to change.

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Agents Comparison

Feature	Simple Reflex	Model-Based Reflex	Goal-Based
Memory	NO	YES	YES
Flexibility	LOW	MEDIUM	HIGH
Works in partially observable?	NO	YES	YES
Uses goals?	NO	NO	YES

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Is a Reflex Agent Purely Rule-Based?

Yes. A reflex agent operates on **predefined, hardcoded rules**.
It follows an **“if condition, then action”** model with **no memory**, **no learning**, and **no planning**.
Example:

if temperature > 37°C → move toward source

It’s **not adaptive**, not goal-oriented, and doesn't optimize its behavior over time.

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Is there any automation in reflex agent?

Not in the intelligent or adaptive sense.

- It **automates reactions**, but it does **not automate decisions** based on goals, learning, or optimization.
- It's basically a glorified **rule engine**.

We can say:

Reflex agents are like vending machines.

Insert input → get output.

But they'll never learn your favorite snack or offer you a deal.

Reflex agent is like a **fan** with a temperature sensor.

It turns on if it's hot. But it doesn't care whether you're in the room or if you're freezing already. It just **reacts**.

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Agents

Reflex Agent = A Hungry Mosquito

"Like a mosquito. It senses warmth & bites. No planning, no hesitation, just pure *if-warm-then-bite* logic." They don't think about consequences or goals. They just react.

Model-Based Reflex Agent = Grandma's TV Remote Memory

"Grandma doesn't see the volume indicator on the TV, but she *remembers* that last time she pressed 'volume up' 5 times. So this time, she only taps twice.That's a model-based reflex agent."

•Model-based reflex agents rely on internal memory, not just current input.

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Agents

- **Goal-Based Agent = You Before an Exam**
- "You want to get an A (**goal**), but you have options: **study**, **sleep**, **cry**, or **watch Netflix**. A goal-based agent chooses the path that leads to the goal, ideally avoiding crying."
- Goal-based agents evaluate possible actions to move toward a goal.

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Self-Driving Car Analogy

- **Reflex Car**: "If red light, stop." But it stops even at a **red Hoarding board** because it can't differentiate.
- **Model-Based Car**: Remembers where intersections are and acts smarter.
- **Goal-Based Car**: Knows you want to reach home. Plans routes, avoids traffic, reroutes intelligently.

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Agentic AI

- Agentic AI = AI that doesn't just react, but plans, adapts, and takes initiative.
- Reflex AI is like that one friend who only answers with **Yes/No**.
- Goal-based AI is like the friend who helps you reach your destination.
  Agentic AI is the **super-smart friend** who says: 'Don't worry, I already booked your Pathao, grabbed snacks, and set the playlist.'

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classical AI Agents vs modern idea of *Agentic A*

- **Goal-Based Agent (Classical AI sense)**
- Works by:
  - Perceiving the environment.
  - Checking actions that help achieve a defined goal.
- Example: **Self-driving car** → its goal is to reach a destination safely.
- Limitation:
  - Still **bounded** → it can only act within pre-programmed logic and goals.
  - Doesn't usually learn or extend itself to new, undefined situations.

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Agentic AI (Modern sense)

- A more recent concept (popularized with LLMs + autonomous systems).
- Goes **beyond just being goal-based**.
- Features:
  - **Autonomy** → can act without constant human instructions.
  - **Planning & Reasoning** → breaks down complex tasks into sub-tasks.
  - **Tool Use** → can call APIs, search the web, run code, interact with other systems.
  - **Adaptation** → learns and changes strategies dynamically.
- Example: **AutoGPT / Devin (AI software engineer)** → not just solving one goal, but breaking it into steps, choosing tools, and executing.

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